

Partial Fractions and Improper Integrals



Partial fraction decomposition: distinct linear factors

- 1 Decompose $\frac{5x-1}{(x-1)(x+2)}$ into partial fractions.
 - 2 Evaluate $\int \frac{3x+5}{x^2+x-2} dx$ using partial fractions (note $x^2+x-2 = (x+2)(x-1)$).
 - 3 Decompose $\frac{x+7}{x^2-x-6}$ into partial fractions (note $x^2-x-6 = (x-3)(x+2)$).
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Partial fractions with a repeated linear factor

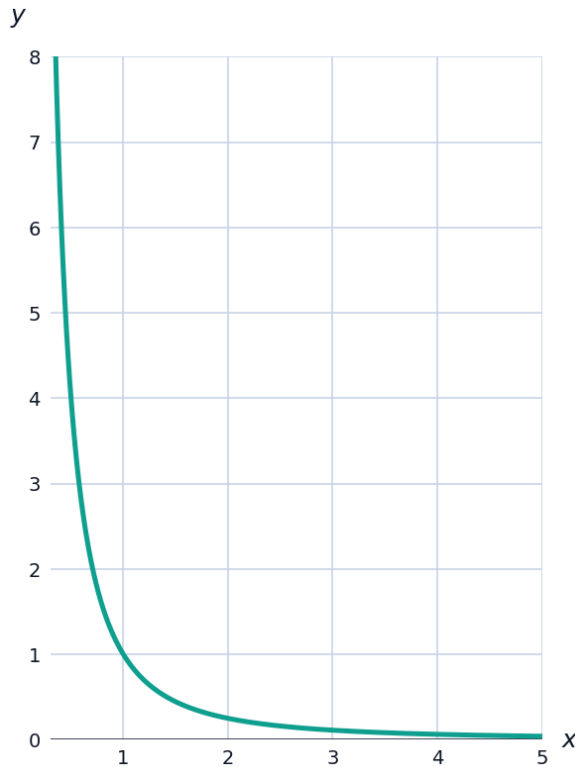
- 4 Decompose $\frac{2x+3}{(x-1)^2}$ into partial fractions.
 - 5 Evaluate $\int \frac{2x+3}{(x-1)^2} dx$ using the decomposition found above.
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Partial fractions with an irreducible quadratic factor

- 6 Set up (but do not solve for constants) the partial fraction decomposition of $\frac{3x^2+2}{(x-1)(x^2+4)}$.
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Improper integrals: infinite limits

- 7 Evaluate $\int_1^{\infty} \frac{1}{x^2} dx$, or show it diverges.



- 8 Evaluate $\int_1^{\infty} \frac{1}{x} dx$, or show it diverges.
- 9 Evaluate $\int_0^{\infty} e^{-2x} dx$, or show it diverges.
- 10 Evaluate $\int_{-\infty}^0 e^x dx$, or show it diverges.

Improper integrals: discontinuous integrands

- 11 Evaluate $\int_0^1 \frac{1}{\sqrt{x}} dx$, or show it diverges (note the integrand is undefined at $x = 0$).
- 12 Evaluate $\int_0^1 \frac{1}{x} dx$, or show it diverges.
- 13 Evaluate $\int_0^2 \frac{1}{(x-2)^2} dx$, or show it diverges (the integrand is undefined at $x = 2$).

The p-test for improper integrals

- 14 State the p-test for $\int_1^{\infty} \frac{1}{x^p} dx$, and use it to determine convergence for $p = 3$ and $p = 0.5$.
- 15 State the analogous p-test for $\int_0^1 \frac{1}{x^p} dx$ (a discontinuity at $x = 0$), and determine convergence for $p = \frac{1}{2}$ and $p = 2$.

Combining partial fractions with improper integrals

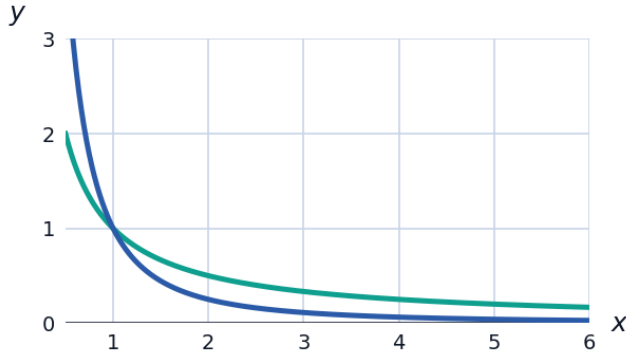
- 16 Evaluate $\int_0^{\infty} \frac{1}{(x+1)(x+2)} dx$ using partial fractions.

Three distinct linear factors

- 17 Decompose $\frac{6x^2 - 3x - 1}{x(x-1)(x+1)}$ into partial fractions.
- 18 Evaluate $\int \frac{6x^2 - 3x - 1}{x(x-1)(x+1)} dx$ using the decomposition above.

Visualizing convergence: area under the curve

- 19 The convergent improper integral $\int_1^{\infty} \frac{1}{x^2} dx = 1$ (shown earlier) means the infinite region under $y = \frac{1}{x^2}$ for $x \geq 1$ has FINITE area. Explain intuitively why this is possible even though the region extends forever.
- 20 By contrast, $\int_1^{\infty} \frac{1}{x} dx$ diverges. Compare the graphs of $y = \frac{1}{x}$ and $y = \frac{1}{x^2}$ on $[1, 10]$ and explain why $\frac{1}{x}$ decays too slowly for the area to stay finite.



Improper integrals with both an infinite bound and a discontinuity

- 21 Evaluate $\int_0^{\infty} \frac{1}{\sqrt{x}(1+x)} dx$ is a classic doubly-improper integral (discontinuous at $x = 0$ AND infinite upper bound). Explain how such an integral must be split to evaluate it.

Comparison test for improper integrals

- 22 Use a direct comparison to argue that $\int_1^{\infty} \frac{1}{x^2 + 1} dx$ converges, without computing it exactly.

23 Use a direct comparison to argue that $\int_1^{\infty} \frac{2 + \sin x}{x} dx$ diverges.